

```
In [1]: import pandas as pd
import numpy as np
```

```
In [2]: df = pd.read_csv('blinkit customer behaviour unclean dataset.csv')
```

to upload file in jupyter notebook

```
In [3]: df.columns.values
```

```
Out[3]: <StringArray>
[      'customer_id',      'customer_name',      'age',
        'gender',          'city',      'subscription',
        'discount_applied', 'item_purchased',      'category',
        'purchase_amount', 'payment_method', 'purchase_date']
Length: 12, dtype: str
```

To see columns values

```
In [4]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 510 entries, 0 to 509
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customer_id            510 non-null    int64
1   customer_name          510 non-null    str
2   age                    510 non-null    int64
3   gender                 510 non-null    str
4   city                   510 non-null    str
5   subscription           510 non-null    str
6   discount_applied       360 non-null    str
7   item_purchased         494 non-null    str
8   category               510 non-null    str
9   purchase_amount        510 non-null    int64
10  payment_method         510 non-null    str
11  purchase_date          510 non-null    str
dtypes: int64(3), str(9)
memory usage: 47.9 KB
```

to see all info of the data

```
In [5]: df.isnull().sum()
```

```
Out[5]: customer_id            0
customer_name            0
age                      0
gender                   0
city                     0
subscription             0
discount_applied        150
item_purchased           16
category                 0
purchase_amount          0
payment_method           0
purchase_date            0
dtype: int64
```

to see the all null values has been there or not

```
In [6]: df["discount_applied"] = df["discount_applied"].fillna("No")
df["item_purchased"] = df["item_purchased"].fillna("Unknown")
```

to remove all null values

```
In [7]: df.isnull().sum()
```

```
Out[7]: customer_id            0
customer_name            0
age                      0
gender                   0
city                     0
subscription             0
discount_applied         0
item_purchased            0
category                 0
purchase_amount          0
payment_method           0
purchase_date            0
dtype: int64
```

```
In [8]: df.duplicated().sum()
```

```
Out[8]: np.int64(10)
```

to find all duplicate raw

```
In [9]: df = df.drop_duplicates()
```

to remove all duplicate raws

```
In [10]: df.duplicated().sum()
```

```
Out[10]: np.int64(0)
```

```
In [11]: df.dtypes
```

```
Out[11]: customer_id      int64
customer_name      str
age      int64
gender      str
city      str
subscription      str
discount_applied      str
item_purchased      str
category      str
purchase_amount      int64
payment_method      str
purchase_date      str
dtype: object
```

to see data type

```
In [25]: df.columns.values
```

```
Out[25]: <StringArray>
[      'customer_id',      'customer_name',      'age',
      'gender',      'city',      'subscription',
      'discount_applied',      'item_purchased',      'category',
      'purchase_amount',      'payment_method',      'purchase_date']
Length: 12, dtype: str
```

```
In [26]: text_columns = ["customer_name", "gender", "city", "subscription",
                        "discount_applied", "item_purchased", "category",
                        "payment_method"]

for columns in text_columns:
    df[columns] = df[columns].str.strip()
```

to remove extra spaces and string value in text

```
In [27]: df.head()
```

```
Out[27]:
```

	customer_id	customer_name	age	gender	city	subscription	discount_applied	item_purchased	categor
0	1001	Sara	50	Male	Gurugram	No	Yes	Face Wash	Beaut
1	1002	Vikas	24	Female	Delhi	No	No	Keyboard	Electronic
2	1003	Rohan	50	Female	Delhi	Yes	No	Watch	Fashio
3	1004	Ali	42	Male	Noida	Yes	No	Tea	Grocerie
4	1005	Karan	56	Male	Meerut	Yes	Yes	Keyboard	Electronic

```
In [28]: df["city"] = df["city"].str.title()
```

```
In [30]: df.columns.values
```

```
Out[30]: <StringArray>
[      'customer_id',      'customer_name',      'age',
      'gender',      'city',      'subscription',
      'discount_applied',      'item_purchased',      'category',
      'purchase_amount',      'payment_method',      'purchase_date']
Length: 12, dtype: str
```

```
In [37]: df["gender"] = df["gender"].str.title()
```

```
In [38]: df["discount_applied"] = df["discount_applied"].str.title()
```

```
In [39]: df["payment_method"] = df["payment_method"].str.title()
```

to standardize the columns

```
In [45]: df[df["purchase_amount"] < 0]
```

Out[45]:

	customer_id	customer_name	age	gender	city	subscription	discount_applied	item_purchased	category	
	14	1015	Neha	24	Female	Meerut	Yes	Yes	Tea	Groceries
	21	1022	Priya	47	Female	Gurugram	No	No	Rice	Groceries
	22	1023	Aman	20	Male	Lucknow	No	No	Laptop	Electronics
	40	1041	Farhan	53	Male	Gurugram	Yes	No	Laptop	Electronics
	52	1053	Karan	33	Female	Noida	Yes	No	Football	Sports
	87	1088	Farhan	50	Female	Lucknow	Yes	No	Cricket Bat	Sports
	142	1143	Rohan	33	Male	Lucknow	Yes	No	Shoes	Fashion
	145	1146	Farhan	48	Female	Gurugram	No	Yes	Face Wash	Beauty
	169	1170	Zoya	27	Male	Gurugram	Yes	No	Football	Sports
	210	1211	Neha	21	Male	Meerut	Yes	Yes	Milk	Groceries
	222	1223	Karan	52	Female	Noida	No	No	Mouse	Electronics
	257	1258	Vikas	27	Male	Delhi	No	Yes	Milk	Groceries
	265	1266	Neha	54	Male	Noida	No	No	Milk	Groceries
	286	1287	Neha	21	Female	Lucknow	No	No	Mobile	Electronics
	290	1291	Karan	24	Female	Delhi	Yes	Yes	Shampoo	Beauty
	301	1302	Aman	47	Female	Delhi	No	Yes	Rice	Groceries
	335	1336	Neha	38	Male	Noida	Yes	No	T-Shirt	Fashion
	347	1348	Aman	23	Male	Noida	No	Yes	Cricket Bat	Sports
	374	1375	Sara	25	Male	Lucknow	No	No	Rice	Groceries
	375	1376	Karan	51	Female	Noida	No	No	Football	Sports
	420	1421	Sara	24	Male	Delhi	No	No	Rice	Groceries
	480	1481	Neha	38	Female	Lucknow	Yes	Yes	Mouse	Electronics

```
In [46]: df["city"].unique()
```

```
Out[46]: <StringArray>
['Gurugram', 'Delhi', 'Noida', 'Meerut', 'Lucknow']
Length: 5, dtype: str
```

```
In [47]: df["gender"].unique()
```

```
Out[47]: <StringArray>
['Male', 'Female']
Length: 2, dtype: str
```

```
In [48]: df["payment_method"].unique()
```

```
Out[48]: <StringArray>
['Net Banking', 'Card', 'Upi', 'Cash']
Length: 4, dtype: str
```

```
In [49]: df.columns.values
```

```
Out[49]: <StringArray>
['customer_id', 'customer_name', 'age', 'gender', 'city', 'subscription', 'discount_applied', 'item_purchased', 'category', 'purchase_amount', 'payment_method', 'purchase_date']
Length: 12, dtype: str
```

```
In [50]: df["subscription"].unique()
```

```
Out[50]: <StringArray>
['No', 'Yes']
Length: 2, dtype: str
```

```
In [51]: df["discount_applied"].unique()
```

```
Out[51]: <StringArray>
['Yes', 'No']
Length: 2, dtype: str
```

```
In [52]: df["category"].unique()
```

```
Out[52]: <StringArray>
['Beauty', 'Electronics', 'Fashion', 'Groceries', 'Sports']
Length: 5, dtype: str
```

```
In [54]: (df["purchase_amount"] < 0).sum()
```

```
Out[54]: np.int64(22)
```

```
In [56]: ((df["purchase_amount"] < 0).sum() / len(df)) * 100
```

```
Out[56]: np.float64(4.3999999999999995)
```

```
In [57]: df["purchase_amount"] = df["purchase_amount"].abs()
```

```
In [60]: (df["purchase_amount"] < 0).sum()
```

```
Out[60]: np.int64(0)
```

```
In [61]: df.describe()
```

```
Out[61]:
```

	customer_id	age	purchase_amount
count	500.000000	500.000000	500.000000
mean	1250.500000	38.320000	36246.914000
std	144.481833	12.65836	20744.026136
min	1001.000000	18.000000	222.000000
25%	1125.750000	27.000000	19356.500000
50%	1250.500000	38.000000	34855.000000
75%	1375.250000	50.000000	55156.000000
max	1500.000000	60.000000	69834.000000

```
In [62]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customer_id           500 non-null   int64
1   customer_name         500 non-null   str
2   age                   500 non-null   int64
3   gender                500 non-null   str
4   city                  500 non-null   str
5   subscription          500 non-null   str
6   discount_applied     500 non-null   str
7   item_purchased        500 non-null   str
8   category              500 non-null   str
9   purchase_amount       500 non-null   int64
10  payment_method        500 non-null   str
11  purchase_date          500 non-null   str
dtypes: int64(3), str(9)
memory usage: 47.0 KB
```

```
In [64]: df["purchase_date"] = pd.to_datetime(df["purchase_date"])
```

to convert text to date format in purchase_date

```
In [65]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customer_id           500 non-null   int64
1   customer_name         500 non-null   str
2   age                   500 non-null   int64
3   gender                500 non-null   str
4   city                  500 non-null   str
5   subscription          500 non-null   str
6   discount_applied      500 non-null   str
7   item_purchased        500 non-null   str
8   category              500 non-null   str
9   purchase_amount       500 non-null   int64
10  payment_method        500 non-null   str
11  purchase_date         500 non-null   datetime64[us]
dtypes: datetime64[us](1), int64(3), str(8)
memory usage: 47.0 KB
```

```
In [72]: df.columns.values
```

```
Out[72]: <StringArray>
[      'customer_id',      'customer_name',      'age',
      'gender',      'city',      'subscription',
      'discount_applied',      'item_purchased',      'category',
      'purchase_amount',      'payment_method',      'purchase_date']
Length: 12, dtype: str
```

```
In [77]: df["age_group"] = pd.cut(
    df["age"],
    bins = [17,25,35,45,60],
    labels=["18-25","26-35","36-45","46-60"]
)
```

to create age group in dataframe

```
In [81]: df['purchase_month'] = df["purchase_date"].dt.month_name()
```

to create a month columns in dataframe

```
In [83]: df["purchase_day"] = df["purchase_date"].dt.day_name()
```

to create a day column in dataframe

```
In [85]: df["purchase_year"] = df["purchase_date"].dt.year
```

to create year column in dataframe

```
In [86]: df.columns
```

```
Out[86]: Index(['customer_id', 'customer_name', 'age', 'gender', 'city', 'subscription',
      'discount_applied', 'item_purchased', 'category', 'purchase_amount',
      'payment_method', 'purchase_date', 'age_group', 'purchase_month',
      'purchase_day', 'purchase_year'],
      dtype='str')
```

```
In [89]: df["discount_flag"] = df["discount_applied"].map({
    "Yes":1,
    "No":0
})
```

to create a discount_flag column in dataframe

```
In [90]: df.head()
```

```
Out[90]:
```

	customer_id	customer_name	age	gender	city	subscription	discount_applied	item_purchased	categor
0	1001	Sara	50	Male	Gurugram	No	Yes	Face Wash	Beaut
1	1002	Vikas	24	Female	Delhi	No	No	Keyboard	Electronic
2	1003	Rohan	50	Female	Delhi	Yes	No	Watch	Fashio
3	1004	Ali	42	Male	Noida	Yes	No	Tea	Grocerie
4	1005	Karan	56	Male	Meerut	Yes	Yes	Keyboard	Electronic

```
In [91]: df["revenue_segment"] = pd.cut(
    df["purchase_amount"],
```

```
    bins = [0,10000,30000,50000,70000],  
    labels = ["low","medium","high","premium"]  
)
```

```
In [92]: df.columns
```

```
Out[92]: Index(['customer_id', 'customer_name', 'age', 'gender', 'city', 'subscription',  
               'discount_applied', 'item_purchased', 'category', 'purchase_amount',  
               'payment_method', 'purchase_date', 'age_group', 'purchase_month',  
               'purchase_day', 'purchase_year', 'discount_flag', 'revenue_segment'],  
              dtype='str')
```

```
In [94]: df.to_csv("blinkit_clean_customer_behaviour.csv", index=False)
```